

Geometric Limits on Semantic Expressibility: Quantum-Inspired Contexts and Polysemy in Large Language Models

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A chosen d -dimensional language-model representation has only d independent coordinates, yet, at fixed angular tolerance, it can accommodate exponentially many nearly orthogonal feature directions. Lévy concentration and Johnson–Lindenstrauss embeddings anchor this claim, while simultaneous readout remains limited by overlap: for uniformly random directions, a typical pair has root-mean-square overlap $1/\sqrt{d}$. We then propose a strictly structural analogy for polysemy. The same lexical anchor ray is retained across different uses, just as one ray may be an element of several orthonormal bases. The surrounding text then controls a separate classical residual map. Thus the complete large language model representation depends on linguistic context, whereas, for a fixed quantum state, the Born probability of the shared projector does not depend on which compatible basis contains it.

I. INTRODUCTION

This chapter examines contextual representation in large language models (LLMs) as part of the volume’s broader inquiry into “quantum thinking.” A short operational picture is enough to begin. Text is divided into tokens; each token is mapped to an initial vector; Transformer layers turn those initially context-free vectors into context-dependent hidden states; and an output map assigns probabilities to possible next tokens [1]. Section II explains this loop in plain language. Appendix A gives an explicit formal version, with every quantity defined, that may be skipped on a first reading.

The central question is simple: how can a space with only d independent coordinates accommodate a much larger dictionary of approximately distinct semantic directions, and what limits their simultaneous recovery by a simple correlation readout? The positive thesis has two parts. High dimension makes a large dictionary geometrically possible, but useful meaning is limited by coactivation and readout. We hypothesize that learning spends its limited angular separation where cross-talk is costly—between important features that often activate together. For lexical ambiguity, a reusable lexical ray may additionally be shared across several context-dependent residual subspaces.

This is an analytical and model-proposal chapter; it reports no new experiment. The geometric results are established mathematics used as null benchmarks. The coactivation prediction and the shared-lexical-ray model are empirical hypotheses, and Sec. VIII D gives tests under which they fail. Stating that status early is important: a possible arrangement of vectors is not evidence that an LLM has learned that arrangement.

a. Elementary mathematical map. Most steps below reduce to familiar facts. The Pythagorean theorem and

the orthogonal projection theorem split a vector into perpendicular components. The Cauchy–Schwarz and triangle inequalities bound inner products and accumulated error. The polarization identity converts squared distances into inner products. The spectral theorem diagonalizes real symmetric matrices and, over complex spaces, normal operators. Finally, additivity of variance explains root-mean-square growth, and the union bound says that the chance of any bad event is at most the sum of their individual chances. The core argument uses these elementary facts. Lévy concentration and the Johnson–Lindenstrauss (JL) lemma are quoted but not derived; a few explicitly optional comparisons use additional standard results and may be skipped.

Let $\mathbb{R}^d$ be the particular representation space used in the geometric analysis. Here d is the number of independent axes retained by the stated measurement protocol; it is the dimension used in the capacity estimates, not necessarily the architectural width of the model. Let M be the measured number of feature directions returned by a declared extraction and duplicate-removal rule. It is a count, not a dimension. For unit vectors, Cauchy–Schwarz gives $|u^\top v| \leq 1$, and $u^\top v$ is the cosine of their angle. We call a dictionary ε -quasi-orthogonal when every absolute pairwise inner product is at most the chosen tolerance $0 < \varepsilon < 1$. Distinct code vectors are then separated by a positive Euclidean distance. By the Heine–Borel theorem the unit sphere is compact, so it has a finite cover by balls smaller than half that separation. Each ball contains at most one code point. Consequently a largest finite code size exists; we denote it by $M_\varepsilon(d)$. It is a resolution-dependent cardinality, not a second dimension [2–4].

High-dimensional concentration and JL projection both lead to the same single-exponential existence scale: for fixed ε and sufficiently large d , one can construct at least $\exp(c\varepsilon^2 d)$ quasi-orthogonal directions for some constant $c > 0$. These are related formulations of one concentration phenomenon, not two independent capacity mechanisms. JL preserves a prescribed finite geometry;

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random spherical coding instead supplies a null distribution for a sampled dictionary and its most extreme overlap. A conditional double exponential after separately assuming a fully accessible tensor-product carrier is discussed separately in the conditional remark in Sec. III C; it is not an LLM capacity claim.

Geometric coexistence is only the first half of the problem. Suppose k random-like features are active together. A simple correlation readout of one chosen feature receives many small cross terms. Their variances add, giving a root-mean-square (RMS) interference scale $\sqrt{k/d}$. This is the Pythagorean rule applied to random errors: it describes average squared fluctuation, not cancellation in every example. The empirical prediction is therefore about the placement of overlap, not merely its existence. Important, frequently coactive features should cause less held-out cross-talk than activity-matched controls.

Lexical ambiguity supplies the second hypothesis. The same token can occur in financial, river-edge, or seat/bench uses of “bank,” while its surrounding sequence drives different contextual states. In an ordinary Transformer, every occurrence of the same token already begins with the same lookup vector; the later hidden vectors are generally different because self-attention and the remaining layers depend on the surrounding sequence. Our proposal makes that distinction into an explicit structural constraint. A fixed unit vector r_t represents the reusable lexical anchor; several partial orthonormal structures contain that same vector, and a classical context-dependent map supplies the remaining sense information. The nontrivial claim is therefore not that a shared input embedding exists. It is that retaining an identifiable shared ray at a chosen representation layer, while conditioning the complementary directions and readout on context, improves held-out prediction, interpretability, or sample efficiency relative to equally expressive untied models.

Current Transformers remain classical systems. Real vectors, inner products, high dimension, or probabilistic output do not make a computation quantum. Here “quantum-inspired” has one deliberately narrow meaning. In quantum theory, a unit vector determines a ray, equivalently a rank-one projector, and that ray may occur as one outcome in more than one orthonormal basis. The bases are then interlocking measurement contexts with a common element. In dimension at least three they can be distinct while sharing only that element. For a fixed quantum state ρ , the Born probability of the shared projector P is $\text{tr}(\rho P)$ and is independent of which other orthogonal projectors complete the basis. Gleason’s theorem, under its hypotheses in dimension at least three, characterizes probability assignments of this basis-independent form [5, 6]. Here ρ is the matrix representing the fixed quantum state, P projects onto one possible outcome ray, and tr means the trace, the sum of a matrix’s diagonal entries. This limited context independence does not mean that the probability is independent of the quantum state.

The proposed language-model encoding copies only the incidence pattern, not the quantum probability rule. The same lexical ray is placed in several partial orthonormal structures, each completable to a basis and drawable as a hyperedge sharing one vertex with the others, but the semantic map is deliberately allowed to depend on the surrounding text. A hypergraph records which elements belong to which contexts; it does not by itself prescribe probabilities, dynamics, or noncommutativity [7]. Thus the analogy is structural, whereas the LLM readout remains explicitly context dependent. The narrow statement about a shared quantum projector should not be confused with claims about state update, sequential measurements, or Kochen–Specker contextuality. The relation to quantum cognition and compositional distributional semantics is reviewed in Sec. VII B; no physical quantumness or semantic Kochen–Specker theorem is inferred.

A practical caveat is that an LLM layer may use fewer directions than its architectural width suggests. Let that width be d_{model} . Suppose we collect n hidden-state vectors $h^{(1)}, \dots, h^{(n)} \in \mathbb{R}^{d_{\text{model}}}$ at one layer and subtract their sample mean $\bar{h}$. Their empirical covariance matrix is

$$C = \frac{1}{n} \sum_{i=1}^n (h^{(i)} - \bar{h})(h^{(i)} - \bar{h})^T. \quad (1)$$

Its diagonal entries measure variation in individual coordinates, its off-diagonal entries measure how pairs of coordinates vary together, and $u^T C u$ is the average squared variation in the unit direction u . The matrix is real and symmetric, so the spectral theorem gives an orthonormal eigenbasis with nonnegative eigenvalues. Very small eigenvalues identify directions that the sample scarcely uses. After declaring a threshold, one may retain those eigenvectors above it and, if desired, rescale them in a whitening step. There can be at most $\min(d_{\text{model}}, n - 1)$ nonzero eigenvalues. In this chapter d counts the retained axes. It therefore depends on the chosen layer, sample, centering, threshold, and whitening rule; all must be reported when a numerical value of d is used.

The lexical proposal is logically independent of the global geometric result. Its suitable data sets, controls, and failure criteria are therefore deferred to Sec. VII D, where they are needed, rather than listed in the introduction.

The remaining sections follow the argument in that order. Section III develops the geometric null model; Sec. IV turns coexistence into a readout problem; Secs. V and VI present the shared lexical ray and its context-dependent router; and Sec. VII relates the proposal to existing work, limitations, and proposed experiments.

For reference, $\mathbb{R}^d$ is the usable Euclidean carrier, $u^T v$ denotes the inner product, the superscript T denotes transpose, and $\|v\| = \sqrt{v^T v}$. The unit sphere is $S^{d-1} = \{v \in \mathbb{R}^d : \|v\| = 1\}$; I is the identity matrix; $\mathbb{P}(A)$ is the probability of an event A , $\mathbb{E}[Y]$ is the average of a

random quantity Y , and $\text{Var}(Y) = \mathbb{E}[(Y - \mathbb{E}[Y])^2]$ is its mean squared fluctuation. The symbols $\approx$, $\sim$, $\lesssim$, and $\gtrsim$ below indicate scaling or approximation at the stated level, not exact equality. Local ranks and error tolerances are introduced only where they are used.

II. FROM TEXT TO WEIGHTS AND BACK TO TEXT: THREE COMPUTATIONAL LOOPS

The word *weight* is used for two different objects in informal accounts of Transformers. A trained model has persistent numerical parameters: its embedding table, query, key, value, output, and feed-forward matrices, normalization parameters, and final vocabulary readout. Self-attention also computes input-dependent coefficients that are often called attention weights. Those coefficients exist only for the current forward pass. They do not become persistent model parameters. Keeping this distinction visible separates the three computational loops below.

A. Creating the weights: tokenization, contextual vectors, and training

Training begins with text, but a neural network does not receive words or sentences as primitive objects. A tokenizer first segments character strings into a finite vocabulary of token types and replaces each occurrence by an integer identifier. Tokens may be whole words, word fragments, punctuation, spaces, or byte-level units. Modern subword schemes are designed to cover rare and previously unseen words without assigning a separate vocabulary entry to every possible word [8, 9]. The tokenizer is normally learned or chosen before the main language-model optimization and is then held fixed during that optimization.

An embedding table associates each token identifier with an initial real vector. At this first lookup stage, two occurrences of the same token use the same table entry. Position information is also supplied, either through added position vectors or through mechanisms such as rotary transformations inside attention; some systems add role or segment information as well. The lookup vector is therefore not yet the fully contextual meaning of that occurrence. Context dependence is created by the successive Transformer layers.

Within one self-attention head, the current vector at each position is mapped by learned matrices to a query, a key, and a value. The query at one position is compared by an inner product with keys at other permitted positions. In the decoder-only autoregressive Transformer considered here, a causal mask excludes future positions during next-token training. Softmax normalization turns the permitted comparisons into nonnegative attention coefficients, and those coefficients sum to one. The update is therefore a weighted average of the

value vectors. Cauchy–Schwarz bounds each raw query–key comparison by the product of the two vector lengths; the learned matrices determine which directions make that comparison large. Multiple heads can learn different patterns of dependence. Residual connections, normalization, and feed-forward blocks then transform the update further. After many layers, the hidden vector for an occurrence of “bank” can differ substantially between financial, river-edge, and seat/bench contexts even though all occurrences began from the same token-table entry [1].

This forward calculation still does not create new persistent weights. The last hidden state at each training position is converted into scores for the next token. The discrepancy between the predicted distribution and the observed next token defines a training loss. Backpropagation differentiates that loss through the output map, feed-forward blocks, attention operations, and embedding table. Mathematically, backpropagation is repeated application of the multivariable chain rule. An optimizer then changes the persistent parameters. Repeating this cycle over many batches creates the trained weight values. In short, self-attention produces contextual activations and differentiable routes through which error signals flow; gradient-based optimization produces the persistent model weights [10].

Does this description require a Hilbert space? Nothing here requires more than finite-dimensional Euclidean inner-product geometry. Every finite-dimensional Euclidean space is complete and may therefore formally be called a real Hilbert space, but completeness plays no computational role and implies no quantum mechanism. The construction needs ordinary real vectors, inner products, matrices, and nonlinear maps; it does not require complex amplitudes, Born probabilities, or quantum hardware. Nor does training assign one exactly orthogonal vector to every token or every meaning. The quasi-orthogonal geometry studied later concerns a declared dictionary of extracted feature directions at a chosen resolution. It is a possible organization of learned semantic distinctions, not an assumption built into tokenization or self-attention.

B. Generating an output: tokenization and autoregressive feedback

Response generation begins with the same tokenizer. The prompt, including any control, role, or formatting tokens exposed to the model, is converted to a token sequence and then to initial vectors. The trained parameters are now normally fixed. A causal forward pass produces contextual hidden states for the prompt, and the final relevant state is mapped to one score for every token in the vocabulary. Softmax converts these scores into a next-token distribution. Greedy decoding chooses its largest entry; sampling may use a temperature or restrict attention to a high-probability subset before drawing a

token.

The selected token is appended to the sequence. The enlarged sequence is then the context for the following prediction, so the generated token can change all subsequent attention patterns and hidden states. This loop is repeated until an end condition is reached. Implementations commonly cache previous keys and values to avoid recomputing unchanged parts of the prefix, but caching changes efficiency rather than the trained parameters. Finally, the tokenizer’s inverse assembly rules turn the generated token identifiers back into a character string [9].

Autoregressive generation is thus already recursive in a precise but limited sense: each output token becomes part of the next input. The context, attention coefficients, and hidden vectors evolve from step to step, while the persistent model weights remain fixed. The standard decoding interface does not require a separately represented complete answer; it emits tokens successively by repeatedly mapping the current vector-valued context to a distribution over the next discrete token.

C. Recursive adaptation, alignment, and hallucination

Three forms of recursion should be separated. First, in autoregressive generation each output token becomes part of the next input, while the model parameters remain fixed. Second, an external memory, retrieval result, tool output, or longer conversation history can enlarge the next input without changing those parameters. Third, genuine online learning can update selected parameters after an interaction. Dynamic evaluation shows that the third loop is technically possible [11], but it is a separate learning rule rather than ordinary frozen-weight inference.

This subsection is orientation only; no later geometric conclusion depends on it. Online learning needs verified feedback and safeguards against drift, forgetting, poisoning, and error reinforcement. Post-training alignment is also a separate intervention: it changes the objective toward selected forms of instruction following, helpfulness, and safety [12]. Alignment and factual accuracy are not opposite ends of one scale. Preference optimization can encourage agreement in some settings [13], while hallucination also occurs without alignment [14].

Whether a particular alignment procedure increases or decreases unsupported claims is therefore a causal question, not a consequence of the word “alignment.” It requires a matched control with the same base model, data exposure, compute, prompt distribution, grounding, and decoding rule. Appendix A 3 gives the optional update equation, the relevant safeguards, and this matched comparison in more detail.

III. HOW HIGH DIMENSION SUPPORTS MANY NEARLY ORTHOGONAL DIRECTIONS

The question is now geometric: at a fixed angular resolution, how does the number of available directions grow with the exact carrier dimension? Here d is the usable carrier rank, whereas M counts the directions in one measured dictionary and may exceed d . For a dictionary

$$W = \{w_1, \dots, w_M\} \subset S^{d-1}, \quad \mu(W) = \max_{i \neq j} |w_i^\top w_j|. \quad (2)$$

For $0 < \varepsilon < 1$, we call this particular dictionary ε -quasi-orthogonal when $\mu(W) \leq \varepsilon$, and write $M_\varepsilon(d)$ for the maximum cardinality achievable in $\mathbb{R}^d$ under this condition. Thus $M_\varepsilon(d)$ and the empirical count M are cardinalities, not additional vector-space dimensions. The term *quasi-dimensionality* will refer only to this resolution-dependent abundance of directions.

a. Operational meaning of a feature direction. In common linear-probe and feature-dictionary analyses, a learned property is represented or detected by a direction w_i : for a hidden state h , the scalar $w_i^\top h$ measures its component along that direction [15–17]. This interpretation depends on the chosen layer, preprocessing, inner product, and extraction method; it does not assert that every semantic property has one unique direction. For unit directions u and v , the two-feature example $h = au + bv$ gives

$$u^\top h = a + bu^\top v.$$

If $u^\top v = 0$, the component bv contributes nothing to the linear correlation readout along u . If $|u^\top v| \leq \varepsilon$, its contribution has magnitude at most $\varepsilon|b|$. Thus, for this chosen linear readout and inner product, orthogonality eliminates pairwise geometric cross-talk and quasi-orthogonality bounds it. This operational statement does not by itself imply that the represented properties are conceptually unrelated, statistically independent in the data, or causally independent inside the model; nor does it guarantee simultaneous recovery when many features are active.

Two elementary facts place this definition. Exactly orthogonal nonzero vectors are linearly independent, so rank-nullity limits an exactly orthogonal family to d members. Quasi-orthogonal vectors need not be independent. Nevertheless, $M_\varepsilon(d)$ is finite when $\varepsilon < 1$: the condition keeps every pair a fixed positive distance apart, while the unit sphere (with opposite signs identified) is compact. A compact set cannot contain infinitely many points separated by the same positive distance. Thus this is an ordinary finite packing problem, not a new linear dimension.

The mathematical ingredients below are established results. Lévy concentration says that near-orthogonality is typical relative to a fixed direction. The JL lemma repackages the same high-dimensional concentration as a finite-set projection theorem. Direct random coding supplies a null ensemble for the largest overlap in a sampled

dictionary. These are related views of one phenomenon, useful for different diagnostics; their rates are not independent effects to be multiplied. None identifies semantic features in an LLM.

A. Lévy concentration: why random directions are nearly perpendicular

In one standard form of Lévy’s lemma, let $f : S^{d-1} \rightarrow \mathbb{R}$ be L -Lipschitz: changing its input by a distance r changes its value by at most Lr . If m_f is a median of f , then, for $t > 0$,

$$\mathbb{P}\{|f - m_f| \geq t\} \leq 2 \exp\left[-c \frac{(d-1)t^2}{L^2}\right] \quad (3)$$

for a universal constant $c > 0$ [18, 19]. The choice of median or mean and the numerical constant vary among formulations; the invariant content is exponential concentration in dimension for every Lipschitz scalar probe at fixed t/L .

Fix $u \in S^{d-1}$ and take $f_u(x) = u^\top x$. This function is 1-Lipschitz because Cauchy–Schwarz gives $|u^\top(x - y)| \leq \|x - y\|$. By rotational symmetry it has median and mean zero. For uniform $x \in S^{d-1}$,

$$\mathbb{E}[u^\top x] = 0, \quad \text{Var}(u^\top x) = \frac{1}{d}. \quad (4)$$

The variance $1/d$ is just Pythagoras plus symmetry. Rotate coordinates so that u is the first basis vector. Since $x_1^2 + \dots + x_d^2 = 1$ on the unit sphere and the coordinates have identical distributions, each has expected square $1/d$. Chebyshev’s inequality would already show that large overlaps become uncommon, but only with a polynomial bound. Lévy’s lemma sharpens this to the exponential spherical-cap estimate

$$\mathbb{P}\{|u^\top x| \geq \varepsilon\} \leq 2 \exp\left[-\frac{(d-1)\varepsilon^2}{2}\right], \quad 0 < \varepsilon < 1. \quad (5)$$

Thus almost every direction lies close to the equator perpendicular to a specified direction, with a typical overlap amplitude of order $d^{-1/2}$.

There is an equivalent angular description. If the sign of a direction is irrelevant, its principal angle from u is $\theta_{\text{pr}} = \arccos|u^\top x|$. The first-order Taylor approximation near zero is $\arccos z \approx \pi/2 - z$, with only cubic and higher corrections, so its typical deviation from $\pi/2$ is of order $d^{-1/2}$. In the independently sampled isotropic null, a randomly chosen pair therefore exhibits *angular homogenization*: its principal angle lies in a band of width proportional to $d^{-1/2}$ near $\pi/2$ at any fixed confidence level. Increasing the dictionary size at fixed d does not narrow this typical-pair distribution; instead it increases the extreme overlap, as quantified below. High dimension creates both the typical angular concentration and room for exponentially large finite codes.

a. Optional complex comparison. For comparison with complex Hilbert-space geometry, the corresponding complex statement is especially sharp. If ϕ, ψ are independent unit vectors drawn uniformly under all unitary rotations (the meaning of “Haar-random”), then $Z = |\langle \phi, \psi \rangle|^2$ has the Beta(1, $d-1$) distribution. Using the same ε for an overlap *amplitude*, rather than changing midway to a squared-overlap tolerance, gives the exact tail

$$\mathbb{P}\{|\langle \phi, \psi \rangle| \geq \varepsilon\} = (1 - \varepsilon^2)^{d-1} \leq \exp[-(d-1)\varepsilon^2]. \quad (6)$$

Unitary symmetry lets us hold ϕ fixed as the first coordinate vector. The squared coordinate magnitudes sum to one, the complex analogue of Pythagoras, and integrating the first-coordinate density gives the displayed tail. Readers who wish to remain in real vector spaces may skip this exact comparison; no later LLM claim depends on it.

The real and complex formulas differ in constants and exact distributions, but both exhibit the same operative exponential scale $\exp[-cd\varepsilon^2]$.

B. Johnson–Lindenstrauss: preserving a finite cloud of points

Lévy’s lemma describes most points on a sphere. The JL lemma gives a related finite-set formulation: a finite cloud can be compressed while all pairwise squared distances change by at most a chosen relative error. The surprising feature is that the required target dimension grows only as the logarithm of the number of points. Specifically, for $0 < \delta < 1$ and a finite set X , there is a linear map Φ into $\mathbb{R}^d$ such that

$$(1 - \delta)\|x - y\|^2 \leq \|\Phi x - \Phi y\|^2 \leq (1 + \delta)\|x - y\|^2, \quad x, y \in X, \quad (7)$$

provided

$$d \geq C_{\text{JL}} \delta^{-2} \log |X| \quad (8)$$

for a universal C_{JL} [20, 21].

To make the connection to quasi-orthogonality explicit, let e_i be the standard coordinate vector with a one in position i and zeros elsewhere, and take

$$X = \{0, e_1, \dots, e_M\} \subset \mathbb{R}^M.$$

Including the origin is important: Eq. (7) then controls both the norms $\|\Phi e_i\|$ and the distances $\|\Phi e_i - \Phi e_j\|$. The real polarization identity gives

$$2(\Phi e_i)^\top (\Phi e_j) = \|\Phi e_i\|^2 + \|\Phi e_j\|^2 - \|\Phi e_i - \Phi e_j\|^2. \quad (9)$$

This identity follows by expanding the three squared norms and cancelling terms. In the original set, $\|e_i\|^2 = 1$ and $\|e_i - e_j\|^2 = 2$. Because the origin was included, JL controls each projected norm as well as each projected

distance. Substituting those bounds into the polarization identity gives $|(\Phi e_i)^\top (\Phi e_j)| \leq 2\delta$, while $\|\Phi e_i\|^2 \geq 1 - \delta$. After normalization, $w_i = \Phi e_i / \|\Phi e_i\|$, one obtains

$$|w_i^\top w_j| \leq \frac{2\delta}{1 - \delta} \quad (i \neq j). \quad (10)$$

Normalization divides by $\|\Phi e_i\| \|\Phi e_j\| \geq 1 - \delta$, which explains the denominator. Taking $\delta = \varepsilon / (2 + \varepsilon)$ makes the right-hand side of Eq. (10) equal to ε . Reading Eq. (8) constructively therefore gives, for every fixed $0 < \varepsilon < 1$ and all sufficiently large d ,

$$M_\varepsilon(d) \geq \exp(c'\varepsilon^2 d), \quad (11)$$

for a universal $c' > 0$. This is an existence lower bound. Sharp lower bounds for worst-case JL dimension reduction [22, 23] do not, by themselves, establish a matching universal upper bound for spherical codes, and no such upper bound is claimed here. Algebraically, the exponential appears simply by solving the logarithmic dimension condition for the number of points: undoing a logarithm requires exponentiation.

C. Direct random coding and the Welch floor

The exponential scale can also be reached directly. Draw the M directions independently from the uniform measure on S^{d-1} . Applying Eq. (5) to all unordered pairs gives

$$\mathbb{P}\{\mu(W) \geq \varepsilon\} \leq M(M-1) \exp\left[-\frac{(d-1)\varepsilon^2}{2}\right]. \quad (12)$$

There are $\binom{M}{2}$ unordered pairs. The union bound says that the probability that at least one pair is bad is no larger than the sum of the individual bad-pair probabilities; the pair events need not be independent. The factor two in Eq. (5) cancels the factor one-half in the number of pairs. Whenever the right-hand side is less than one, at least one ε -quasi-orthogonal dictionary must exist: failure cannot occur for every random draw. This is the elementary probabilistic method. It reproduces the $\exp(c\varepsilon^2 d)$ lower scale through the same underlying concentration mechanism as JL, now as an ensemble calculation rather than a finite-metric projection statement. The two rates must not be multiplied or counted as independent amplifications. Equation (12) is only a sufficient random-code bound; it is neither an optimized-code upper bound nor a semantic capacity theorem.

Equating the number of pairs to the real spherical-cap tail gives the useful finite-sample null estimate

$$\mu_{\max}^{\text{null}}(d, M) \approx 2\sqrt{\frac{\log M}{d}}, \quad \log M \ll d, \quad (13)$$

up to lower-order spherical-cap factors. The estimate comes from setting the right-hand side of the union

bound to a number of order one. Taking logarithms turns a product into a sum, and solving $M^2 \exp(-d\varepsilon^2/2) \approx 1$ for ε gives the displayed square root. It is therefore an extreme-value calibration for a random ensemble. It describes the most extreme pair in an independently sampled isotropic dictionary, not the overlap of a typical pair and not a hard limit on a learned one.

A genuine deterministic restriction is supplied by the Welch bound:

$$\mu(W)^2 \geq \frac{M-d}{d(M-1)} \quad (M > d), \quad (14)$$

for every unit-vector dictionary [24]. The Welch floor, the random maximum in Eq. (13), and the existence lower bound in Eq. (11) answer different questions and should not be presented as the two edges of one universal interval.

The Welch bound itself follows from familiar linear algebra. Form the Gram matrix $G = (w_i^\top w_j)_{i,j}$, whose entries are all pairwise inner products. It is positive semidefinite, has rank at most d , and has trace M . By the spectral theorem it has at most d nonzero eigenvalues, and Cauchy–Schwarz says that the sum of their squares is at least M^2/d . The diagonal of G already contributes M to that sum of squares. The average squared off-diagonal entry must therefore be at least the right-hand side of Eq. (14); the maximum cannot be smaller than the average.

a. Optional complex code. The exact complex-Haar law in Eq. (6) makes the corresponding random-code statement particularly transparent. If $M_\varepsilon^{\mathbb{C}}(d)$ denotes the maximum cardinality in $\mathbb{C}^d$ at the same amplitude tolerance, a union bound over all pairs gives the explicit existence result

$$M_\varepsilon^{\mathbb{C}}(d) \geq \left\lceil \exp\left[\frac{d-1}{2}(-\log(1-\varepsilon^2))\right] \right\rceil. \quad (15)$$

This is the sharper complex analogue of the real exponential scale, not an additional linear dimension.

b. Conditional tensor-product remark (not an LLM capacity claim). The conditional double exponential requires two separate steps. If an independently justified carrier is the full tensor product $\bigotimes_{a=1}^n \mathbb{R}^q$, with fixed $q > 1$, then its exact dimension is

$$d = q^n. \quad (16)$$

The elementary tensor-product basis is indexed by n independent choices, each with q possibilities, so the multiplication principle gives q^n basis vectors. Quasi-orthogonal coding is then exponential in that already exponential dimension. For fixed $0 < \varepsilon < 1$, substitution into Eq. (11) gives

$$M_\varepsilon(q^n) \geq \exp(c'\varepsilon^2 q^n), \quad (17)$$

which is doubly exponential in n . Equation (15) gives the same conclusion, with an explicit exponent, for a full

complex tensor-product carrier. These bounds concern arbitrary directions in the full carrier. They do not follow if admissible states are restricted to product vectors or another low-complexity manifold; access to the full carrier, or a code construction within the admissible class, is an additional premise. An ordinary Transformer residual stream is presented only as a measured real carrier $\mathbb{R}^d$. Its width, layers, heads, or token positions do not by themselves establish independent semantic tensor factors. For an LLM the unconditional geometric statement is therefore an existence lower bound exponential in the declared usable d ; Eq. (17) is a further tensor-product hypothesis, not an architectural consequence. No later semantic conclusion in this chapter depends on this remark.

IV. FROM GEOMETRIC COEXISTENCE TO SEMANTIC READOUT

An exponential dictionary can coexist geometrically without all its coefficients being simultaneously accessible. Readout depends on which directions activate together, their amplitudes, the decoder, and the required error probability. This distinction is where the geometry acquires potential semantic content.

A. Fixed-target, worst-case, and joint readout

Let an active set S of size k produce the hidden state

$$h = \sum_{j \in S} x_j w_j. \quad (18)$$

Here S is the set of features active in this example, $k = |S|$, and x_j is the amplitude of feature j . This is simply a linear combination of the active directions. For one specified active target $i \in S$, a correlation decoder returns

$$w_i^\top h = x_i + \sum_{\substack{j \in S \\ j \neq i}} x_j w_i^\top w_j. \quad (19)$$

This identity uses only distributivity of the inner product and $w_i^\top w_i = 1$. The first term is the desired signal; every cross inner product in the remaining sum is interference. In the exact independently sampled isotropic ensemble, provided S and the amplitudes are fixed or selected independently of the direction draws, conditional on S , (x_j) , and w_i , the variables $w_i^\top w_j$ for $j \neq i$ are independent, centered, and have variance $1/d$. Consequently

$$\text{Var} \left[\sum_{\substack{j \in S \\ j \neq i}} x_j w_i^\top w_j \middle| S, (x_j), w_i \right] = \frac{1}{d} \sum_{\substack{j \in S \\ j \neq i}} x_j^2. \quad (20)$$

For independent centered random terms, covariance terms vanish, so the variance of a sum is the sum of the

variances. This is the probabilistic version of Pythagoras. It describes average squared fluctuation and does not assert cancellation in each individual context. For a learned, merely random-like dictionary this equality is used only as an approximate null calibration. For comparable order-one amplitudes it gives the fixed-target RMS scale

$$\sigma_{\text{int}} \sim \sqrt{\frac{k-1}{d}} \approx \sqrt{\frac{k}{d}}. \quad (21)$$

This estimate is a property of the declared correlation decoder under cancellation assumptions, not a universal recovery law.

Without cancellation, coherence yields the deterministic target-wise bound

$$|w_i^\top h - x_i| \leq \mu(W) \sum_{\substack{j \in S \\ j \neq i}} |x_j|. \quad (22)$$

This bound follows from the triangle inequality and $|w_i^\top w_j| \leq \mu(W)$. Unlike the RMS estimate, it assumes no cancellation, so the possible error grows linearly with the absolute amplitudes. The distinction becomes sharper for joint recovery. With W_S the matrix whose columns are the active directions, let x_S collect their amplitudes and let $\hat{x}_S$ collect their correlation readouts. With $G_S = W_S^\top W_S$, simultaneous readout gives

$$\hat{x}_S = W_S^\top h = G_S x_S, \quad \|G_S - I\|_{\text{op}} \leq (k-1)\mu(W), \quad (23)$$

where $\|A\|_{\text{op}}$ is the operator norm, the largest factor by which A can stretch a unit vector. The last inequality follows, conservatively, from Gershgorin's theorem. The Gram matrix has ones on its diagonal and pairwise overlaps off the diagonal. Gershgorin's theorem says that each eigenvalue lies within an off-diagonal row sum of a diagonal entry; here that row sum is at most $(k-1)\mu(W)$. If it is below one, G_S is invertible and coefficients on a known support can be recovered by applying G_S^{-1} . If $k > d$, rank-nullity instead makes G_S singular. Support identification must additionally control false readouts among the $M - k$ inactive directions, introducing a multiple-comparison cost. Uniform recovery of every sparse coefficient vector is therefore a stronger problem than the fixed-target calculation above: it requires additional assumptions and is not implied by the coherence or JL statements. We do not use such a uniform-recovery theorem here. The exponential coexistence of directions, fixed-target RMS access, worst-case access, and joint coefficient recovery are four distinct statements.

B. Why semantic interference must be conditioned on coactivation

Maximum coherence assigns the same cost to every pair. A semantic decoder does not: two directions that never activate together can overlap without contributing

to the error in Eq. (19). Conversely, even a moderate overlap can be expensive when the pair coactivates often, carries large amplitudes, and matters to the task. Define the ordered, nonnegative activity-and-importance weight

$$a_{ij} := \omega_i \mathbb{E}[\mathbf{1}_{\{i,j \text{ active}\}} x_j^2], \quad \omega_i \geq 0, \quad (24)$$

where ω_i is a nonnegative scalar recording the task importance of accurately reading feature i . The indicator $\mathbf{1}$ equals one when both features are active and zero otherwise. Thus a_{ij} is the importance of reading i , multiplied by the average squared amplitude of j on examples where the pair coactivates; it vanishes when they never coactivate. For a fixed target i , write the contribution from feature j as $c_{ij} = \mathbf{1}_{\{i,j \text{ active}\}} x_j (w_i^\top w_j)$. If distinct contributions have zero cross-moments on the held-out distribution, $\mathbb{E}[c_{ij} c_{ij'}] = 0$ for $j \neq j'$, then the diagonal part of the expected importance-weighted squared error is

$$E_{\text{int}} = \sum_{i \neq j} a_{ij} (w_i^\top w_j)^2. \quad (25)$$

Squaring the interference sum produces diagonal terms with the same feature index and cross terms with two different indices. Equation (25) keeps the diagonal part; the cross terms vanish only under the stated zero-cross-moment assumption. We use E_{int} as a coactivation-weighted diagonal interference proxy. Without that assumption, the full squared error also contains cross-moment terms involving distinct j and j' . If the contributions are centered, these cross-moments are covariances.

This changes the scientific prediction. A useful learned dictionary need not beat the isotropic null on every pair, and a nonzero overlap is not itself a defect. If training is pressured to reduce semantic readout error, then important, frequently coactive pairs should contribute less to Eq. (25) than they do in controls matched for d , M , marginal activity, amplitude, and feature importance. That conditional redistribution, not overcompleteness alone, is testable.

This prediction extends rather than replaces earlier toy-model work on superposition, where correlated and anticorrelated features can favor different learned geometries [15]. The additional proposal here is the held-out, task-weighted statistic in Eq. (25) for extracted contextual features. The general observation that feature correlation affects packing is not claimed as new.

C. A task-dependent operating window

There is no task-independent numerical bracket whose lower edge is d and whose upper edge is a universal packing capacity. A meaningful operating window is instead the intersection of an empirical coverage requirement with declared readout tolerances. Let $\varepsilon_{\text{task}}$ be the largest

tolerated absolute pairwise overlap for a chosen diagnostic and let Δ be the tolerated fixed-target RMS interference. For the real isotropic null, Eqs. (13) and (21) give the separate crossover scales. They come from elementary algebra: set the random maximum $2\sqrt{\log M/d}$ equal to $\varepsilon_{\text{task}}$, and set the RMS interference $\sqrt{k/d}$ equal to Δ . Solving for the two counts gives

$$M_{\text{crowd}}(d, \varepsilon_{\text{task}}) \approx \exp\left(\frac{d\varepsilon_{\text{task}}^2}{4}\right), \quad k_{\text{rms}}(d, \Delta) \lesssim d\Delta^2. \quad (26)$$

The first approximation assumes the same small-overlap regime used in Eq. (13). The symbols $\approx$ and $\lesssim$ mark design scales, not sharp universal inequalities. The first concerns the worst of order M^2 pairs formed from M independently sampled directions; the second concerns one specified target among k active directions. Let $M_{\text{req}}(\mathcal{T})$ be the smallest measured dictionary size that reaches a prespecified coverage or performance level on task $\mathcal{T}$, and let $E_{\text{max}}(\mathcal{T})$ be its allowed value of the interference proxy. A useful task-conditioned summary is

$$\begin{aligned} M_{\text{req}}(\mathcal{T}) &\leq M, & \mu(W) &\leq \varepsilon_{\text{task}}, \\ k &\lesssim k_{\text{rms}}(d, \Delta), & E_{\text{int}} &\leq E_{\text{max}}(\mathcal{T}). \end{aligned} \quad (27)$$

Read from left to right, the display says: retain enough directions to cover the task; require the measured worst pairwise overlap to stay below its chosen tolerance; activate only as many directions as the RMS budget permits; and keep measured coactivation-weighted error below its task budget. Only the coverage threshold comes from semantic performance. The others are declared geometric or readout criteria. The scale M_{crowd} in Eq. (26) remains an isotropic-null diagnostic, not a substitute for measuring $\mu(W)$ and not a universal spherical-code upper bound. A learned dictionary can lie beyond that random scale and still satisfy the declared overlap tolerance. A stricter joint-recovery criterion may replace the fixed-target condition through Eq. (23). The Goldilocks region is thus the set of dictionaries that provide enough measured task coverage while satisfying explicitly selected pairwise, active-set, and readout-error criteria. Its lower edge comes from the task's coverage curve, not from an orthogonality count.

For scale, if $d = 4096$ and $M = 10^4$, Eq. (13) predicts a random-dictionary maximum of about 9.5%, although one typical pair has RMS overlap only $1/\sqrt{d} \approx 1.6\%$. Separately, $k = 64$ gives the fixed-target scale $\sqrt{k/d} = 0.125$. These numbers illustrate why the dictionary count, the most extreme pair, and simultaneous readout cannot be collapsed into one effective dimension.

Nothing in these benchmarks is specifically semantic until a feature extractor, activity distribution, decoder, and task are fixed. The next section adds a separate and stronger hypothesis about how context-dependent readout may be organized for lexically ambiguous tokens.

V. A SHARED-ANCHOR MODEL OF LEXICAL AMBIGUITY

The previous section supplied classical geometric benchmarks. It did not imply any particular organization of lexical ambiguity. We now propose a separate classical model with a deliberately narrow quantum-logic inspiration. One fixed lexical ray is contained in several partial orthonormal structures, each of which can be completed to an orthonormal basis. The surrounding text selects or mixes context-specific residual directions, so the complete contextual representation generally changes even though its lexical anchor does not. The shared part can represent lexical identity, morphology, or a token-specific baseline. Tying it across contexts pools training examples and reduces the number of separately estimated directions. The expected advantage is regularization, interpretability, or sample efficiency, not greater expressivity. It should be largest for rare or imbalanced uses and can disappear or reverse when no stable shared component exists.

This supplies a possible bridge from global quasi-dimensionality to local meaning. A linguistic context does not decode the entire feature dictionary; it selects a small active set or low-dimensional subspace. The model below organizes that selection by sharing a token-associated ray while placing context-specific content in perpendicular residual subspaces. The global code size and this local organization are independent claims.

Fix one layer ℓ and one centered or whitened carrier for each test. The observed surrounding sequence is x , while s indexes a candidate latent linguistic context class. In supervised experiments a class may be aligned with a labeled word sense; in unsupervised experiments it is latent. It must be inferred from x rather than supplied by an oracle at test time. We abbreviate $h_\ell(t \mid x)$ by $h(x)$ when t and ℓ are fixed. The notation distinguishes mathematical types and roles:

- $r_t \in \mathbb{R}^d$ is a fixed unit lexical anchor;
- $\mathcal{R}_t = \text{span}\{r_t\}$ is its ray, represented without sign ambiguity by the rank-one projector $P_t = r_t r_t^\top$;
- $S_{t,s} \subset \mathcal{R}_t^\perp$ is the low-dimensional residual subspace for linguistic context class s ;
- $u_{t,s,j} \in S_{t,s}$ are orthonormal basis vectors for that subspace;
- $h_\ell(t \mid x) \in \mathbb{R}^d$ is an observed contextual hidden state.

Thus r_t , $u_{t,s,j}$, and $h_\ell(t \mid x)$ are vectors, whereas $\mathcal{R}_t$ and $S_{t,s}$ are subspaces and P_t is a matrix. A quantum pure state is physically a ray rather than a particular unit representative, which is why incidence statements below use P_t . The w_i used in Sec. III have a separate role: they are generic feature directions in the geometric null model and are not reused as sense labels. This section

resets the symbol x : it now denotes sequence context, not the random point on a sphere used in Sec. III A.

Since $\mathcal{R}_t$ has rank one, $\mathcal{R}_t^\perp = \ker(r_t^\top)$ has dimension $d - 1$ by the rank-nullity theorem. Thus every residual basis vector is perpendicular to the lexical anchor; no new ambient dimension is being introduced.

A. The shared lexical-ray hypothesis

Let r_t be a candidate unit lexical direction and set

$$\mathcal{R}_t = \text{span}\{r_t\}, \quad P_t = r_t r_t^\top. \quad (28)$$

Empirically, r_t could be a normalized mean contextual embedding, a learned token-specific probe direction, or a static input embedding after an explicit map into the same layer carrier. It is a hypothesized context-stable lexical anchor, not a mechanical or dynamical object. Any state can be decomposed relative to any direction, so the decomposition alone has no empirical content. This is the orthogonal projection theorem: for every chosen subspace, each vector has a unique closest component inside the subspace and a perpendicular residual. A weak “token component exists” test is also likely to pass simply because contextual states retain token identity. The substantive claim is that a ray fixed from training data can be retained across context classes while context-dependent residual coordinates add held-out predictive value beyond token means, principal-component removal, and matched control models.

For each latent linguistic context class s , define a low-dimensional residual subspace of rank m_s

$$\begin{aligned} S_{t,s} &= \text{span}\{u_{t,s,1}, \dots, u_{t,s,m_s}\}, \\ S_{t,s} &\subset \mathcal{R}_t^\perp, \quad u_{t,s,i}^\top u_{t,s,j} = \delta_{ij}, \\ m_s &\ll d - 1. \end{aligned} \quad (29)$$

Here δ_{ij} equals one when $i = j$ and zero otherwise. Thus $\{r_t, u_{t,s,1}, \dots, u_{t,s,m_s}\}$ is an orthonormal partial frame. The orthonormal-extension theorem—obtainable by continuing the Gram–Schmidt process—says that it can be completed to an orthonormal basis of $\mathbb{R}^d$. If every basis vector is replaced by its rank-one projector, that completed basis is a context hyperedge containing the common vertex P_t . Different context hyperedges share P_t but generally have different remaining vertices. Constructing such completions is automatic linear algebra; the empirical hypothesis concerns whether this tied organization improves a declared held-out task.

For the intended three-arm example, finance, river-edge, and seat/bench associations of “bank” correspond to different subspaces $S_{\text{bank},s}$ inside the common orthogonal carrier $\mathcal{R}_{\text{bank}}^\perp$. The verbal use in which an aircraft banks or tilts supplies a fourth sense and would add a fourth sense subspace; it is mentioned here but omitted from the drawing for clarity. A word sense is not a single vector: topic, syntax, selectional preferences, and world knowledge may require several spanning directions $u_{t,s,j}$.

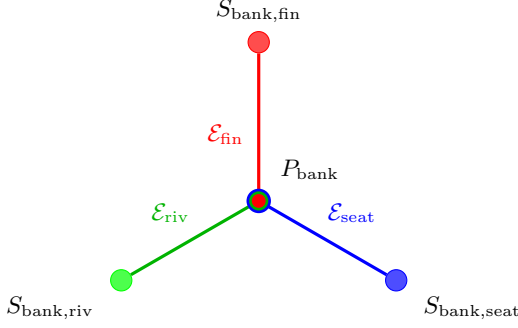


FIG. 1. Structural incidence schematic for the lexically ambiguous token “bank.” The center is the shared lexical projector P_{bank} ; the outer nodes represent displayed residual sectors for the financial, river-edge, and seat/bench contexts. Each colored arm is only the visible fragment of a context hyperedge; omitted mutually orthogonal vertices complete its basis. When $m_s > 1$, an outer $S_{t,s}$ node compresses several rank-one projector vertices associated with an orthonormal basis of that residual subspace; it is not itself one rank-one quantum outcome. The concentric colors mark membership of the same projector in all three contexts. The diagram records shared incidence only, not a vector, map, causal arrow, Born-rule measurement, or claim about what an existing Transformer has learned.

The rank-one construction is the model studied here because it expresses the idea of one and the same lexical ray in several completed bases. A possible extension could share a rank- p subspace

$$\mathcal{R}_t = \text{span}\{r_{t,1}, \dots, r_{t,p}\}, \quad p > 1, \quad (30)$$

with $S_{t,s} \subset \mathcal{R}_t^\perp$. Sharing a subspace does not by itself select particular rays inside it; sharing p specified rank-one projectors is a stronger statement. We therefore use $p = 1$ below and treat higher rank only as an optional modeling extension. Finite quantum-logical structures with more than one shared element are known [25].

The low-rank assumption is empirical. Tying r_t across context classes prevents each class-specific model from independently relearning the proposed common component. This predicts steeper learning curves when examples per use are scarce and a shared component is real; an unconstrained model should catch up with abundant data, and tying may hurt if the assumption is false. It also predicts that dominant within-context variation is low-dimensional after projection onto $\mathcal{R}_t^\perp$. This is compatible with evidence that contextualized embeddings can separate word senses, but it is stronger: it asks whether such separation is privileged relative to a token-associated reference ray [26, 27].

Figure 1 illustrates the shared-incidence structure. Its label $\mathcal{E}_s$ denotes the completed projector family, or context hyperedge, for linguistic context class s ; the drawing shows only the part that contains the common projector and the indicated residual sector.

B. Shared-anchor residual subspaces and classical filters

Collect the residual basis vectors in $B_{t,s} = [u_{t,s,1} \cdots u_{t,s,m_s}]$. Equation (29) says that $B_{t,s}^\top B_{t,s} = I$ and $B_{t,s}^\top r_t = 0$. Thus $B_{t,s} B_{t,s}^\top$ is the orthogonal projector onto $S_{t,s}$, while $P_t = r_t r_t^\top$ projects onto the fixed lexical ray.

The low-rank classical filter associated with context class s is

$$\Pi_{t,s} = P_t + B_{t,s} B_{t,s}^\top. \quad (31)$$

Because the two bases are orthonormal and perpendicular, their cross products vanish. Squaring $\Pi_{t,s}$ therefore returns $\Pi_{t,s}$ itself. This idempotence is the defining property of the orthogonal projector onto the direct sum $\mathcal{R}_t \oplus S_{t,s}$. All $\Pi_{t,s}$ preserve the same direction r_t and differ only in the displayed residual sectors. They are partial filters, not complete quantum measurement contexts: further mutually orthogonal directions must be added to complete the basis described above. No Born probability is assigned to these filters.

Their ordinary matrix action is concrete:

$$h_{t,s}^{(\Pi)}(x) = \Pi_{t,s} h(x) = P_t h(x) + B_{t,s} B_{t,s}^\top (I - P_t) h(x). \quad (32)$$

The first term retains the component along the lexical anchor; the second retains only the residual component selected by context class s . One may select a single s or take a context-dependent soft mixture as in Sec. VI. Equations (31) and (32) define a proposed constrained module or probe; they do not claim that current LLMs already implement it. Quantum-logic discussions sometimes call contexts with shared outcomes *intertwining* or *interlocking*. Here that word refers only to the shared hypergraph incidence, not to an operator equation or physical quantum process. Everything here uses ordinary real matrices on classical hardware.

C. Projection energy and context-class readout

Let $h(x) \in \mathbb{R}^d$ be a contextual state and retain the orthogonal projector P_t defined above. The two components are

$$\begin{aligned} h(x) &= h_{\text{ref}}(x) + h_\perp(x), \\ h_{\text{ref}}(x) &= P_t h(x), \\ h_\perp(x) &= (I - P_t) h(x) \in \mathcal{R}_t^\perp. \end{aligned} \quad (33)$$

The orthogonal projection theorem makes this decomposition unique: $P_t h$ is the closest vector in $\mathcal{R}_t$ to h , and $(I - P_t)h$ is perpendicular to that subspace. Pythagoras gives $\|h\|^2 = \|P_t h\|^2 + \|(I - P_t)h\|^2$. The hypothesis is that most sense-disambiguating variation is present in $h_\perp(x)$, not that the decomposition itself is special. A

basis-invariant score can be expressed directly with the orthonormal matrices already defined:

$$\begin{aligned} e_t(x) &= |r_t^\top h(x)|^2 = \|P_t h(x)\|^2, \\ \beta_{t,s}(x) &= B_{t,s}^\top h_\perp(x), \\ f_{t,s}(x) &= \|\beta_{t,s}(x)\|^2 = \|B_{t,s} B_{t,s}^\top h_\perp(x)\|^2. \end{aligned} \quad (34)$$

The scalar $r_t^\top h$ and the entries of $B_{t,s}^\top h_\perp$ are ordinary coordinates in orthonormal frames. Parseval’s identity says that the sum of their squared coordinates equals the squared length of the corresponding orthogonal projection. This is why the energies do not depend on the particular orthonormal basis chosen inside the same subspace. Here e_t measures energy along the candidate lexical ray and $f_{t,s}$ measures projection energy in the candidate residual subspace. The latter is invariant under changes of orthonormal basis within the same residual subspace. An illustrative projection-energy score is

$$\gamma_{t,s}(x) = \frac{f_{t,s}(x)}{m_s}. \quad (35)$$

The division by the subspace rank matters when the m_s differ: for an isotropic residual, expected raw projection energy is proportional to rank. With equal ranks it does not change the ordering. A more careful empirical implementation may instead calibrate each raw energy against its own held-out null distribution. The selected context-class label is

$$s^*(x) = \arg \max_s \gamma_{t,s}(x). \quad (36)$$

Multiplying every $\gamma_{t,s}$ by the same positive function of e_t would not change this ranking; reference energy can instead supply an independently calibrated engagement or abstention threshold. The score is an illustrative probe, not a derivation of how an LLM must disambiguate. At test time the true sense label is unavailable, so a practical implementation must learn the router from context without oracle access. The empirical content is the proposed separation of roles: the anchor component tracks token identity or engagement, while the orthogonal component contains sense-disambiguating information. If no token-associated $\mathcal{R}_t$ exposes sense structure better than centered, token-mean, frequency-controlled, layer-matched, and random controls, the hypothesis fails.

D. Relation to dynamic contextual embeddings

Transformers already handle lexical ambiguity dynamically. Every occurrence of token type t begins with the same embedding-table row, apart from position information, but its later hidden state $h_\ell(t | x)$ depends on the surrounding sequence x . It is therefore correct to say that a modern LLM uses one initial token vector and generally different contextual occurrence vectors. The shared-anchor proposal is a constraint on how those two levels are related, not an alternative hardware architecture.

Its strict form preserves the same token-associated vector $a_t r_t$ at a chosen representation layer:

$$\begin{aligned} \tilde{h}_\ell(t | x) &= a_t r_t + z_\ell(t, x), \\ r_t^\top z_\ell(t, x) &= 0. \end{aligned} \quad (37)$$

Here a_t is learned for token type t but does not depend on x ; hence $a_t r_t$ is literally the same lexical component in every use. The residual $z_\ell(t, x)$ depends on the surrounding text and carries the contextual variation. For an unconstrained pretrained model, the identity $h = P_t h + (I - P_t)h$ is automatic, but constancy of the coefficient $r_t^\top h_\ell(t | x)$ is not. One can therefore test a weaker diagnostic version by measuring that coefficient’s held-out variation, or enforce the strict version in a fitted adapter.

The hypothesis predicts that the residual carries useful disambiguating information and lies near the low-dimensional subspace $S_{t,s}$ selected by the context. Labeled senses should be more separable in $\mathcal{R}_t^\perp$ than in complements of matched control rays.

Thus the critical question is not whether contextual embeddings move; they do. It is whether the same lexical anchor can be retained while their sense-relevant residual movement is organized by residual subspaces, and whether that constraint predicts held-out data better than matched untied or unstructured models.

VI. ATTENTION AND LATENT CONTEXT SELECTION

A standard attention head computes

$$h' = \sum_j \alpha_j v_j, \quad \alpha_j = \frac{\exp(q_{\text{att}}^\top k_{\text{att},j} / \sqrt{d_{\text{head}}})}{\sum_{j'} \exp(q_{\text{att}}^\top k_{\text{att},j'} / \sqrt{d_{\text{head}}})}. \quad (38)$$

$q_{\text{att}}, k_{\text{att},j} \in \mathbb{R}^{d_{\text{head}}}$ are the query and key vectors of that head, and v_j is its value vector. The local head width d_{head} is an architectural quantity, not necessarily the selected carrier dimension d used in the geometric null model. Keys determine the attention coefficients, while values are the vectors being averaged. Because they are produced by different learned maps, ordinary attention is routing rather than orthogonal projection or basis reconstruction.

Softmax makes every α_j nonnegative and makes the coefficients sum to one, so h' is a convex combination of the value vectors. Cauchy–Schwarz bounds a raw query–key score by the product of their lengths. The factor $1/\sqrt{d_{\text{head}}}$ keeps typical score fluctuations of order one when many roughly independent coordinate contributions are added.

The useful analogy is therefore limited: attention amplifies some contextual directions and suppresses others. In the terminology of Sec. V, this resembles context-dependent subspace selection, but it is not the orthogonal projection used in the geometric model.

A usable context selector cannot receive the true sense label at inference. Let learned router logits $\ell_{\theta,s}(x,t)$, with router parameters θ , depend only on the available sequence and token state, and define

$$p_{\theta}(s | x, t) = \frac{\exp(\ell_{\theta,s}(x, t)/\tau_r)}{\sum_{s'} \exp(\ell_{\theta,s'}(x, t)/\tau_r)}, \quad \tau_r > 0. \quad (39)$$

Here τ_r is the router temperature: larger values flatten the context-class probabilities and smaller values sharpen them. These are ordinary classical conditional probabilities learned from linguistic data. They are not Born probabilities and are not constrained by Gleason’s theorem. The router may use the illustrative score in Eq. (35), but need not do so. To write the strict shared-anchor code explicitly, let $z_{t,s}(x) \in \mathbb{R}^{m_s}$ be residual coordinates produced from the sequence. Then

$$h_t^{\text{prop}}(x) = a_t r_t + \sum_s p_{\theta}(s | x, t) B_{t,s} z_{t,s}(x). \quad (40)$$

Require $a_t \neq 0$ and choose a nondegeneracy criterion on training data so that the anchor carries measurable token information rather than collapsing to a negligible component. The first term is independent of x ; the router weights, residual coordinates, and therefore the complete representation are context dependent. Hard routing keeps only the residual subspace with the largest probability. A diagnostic that filters an already observed hidden state instead uses

$$\bar{h}_t(x) = \sum_s p_{\theta}(s | x, t) \Pi_{t,s} h(x). \quad (41)$$

The effective operator $\sum_s p_{\theta}(s | x, t) \Pi_{t,s}$ is generally not idempotent and hence is not itself an orthogonal projector. Squaring a weighted sum creates cross terms, so the result need not equal the original sum. Attention may supply its routing signal, but neither the shared anchor nor the subspace structure follows from standard attention. The constrained model must therefore be compared with parameter-matched mixture-of-experts and unstructured low-rank alternatives.

For an implementable adapter at a chosen layer, let $Z_{t,s} \in \mathbb{R}^{d \times m_s}$ be a trainable seed matrix such that $(I - P_t)Z_{t,s}$ has full column rank, and set $B_{t,s} = \text{orth}((I - P_t)Z_{t,s})$, where orth means ordinary Gram–Schmidt orthonormalization or an equivalent matrix factorization. Multiplication by $I - P_t$ first removes every component along the shared ray; full column rank ensures that m_s independent residual directions remain. Tie the same r_t across all s , compute the router, and obtain $\bar{h}_t$ from Eq. (41). The filtered state may be used as a diagnostic probe. A strict shared-anchor adapter instead forms the post-adapter state

$$h_t^+(x) = a_t r_t + (I - P_t)[h(x) + \eta U \bar{h}_t(x)], \quad (42)$$

where $U \in \mathbb{R}^{d \times d}$ is a learned output map and η is a scalar gate. Multiplication by $I - P_t$ ensures $P_t h_t^+(x) =$

$a_t r_t$ for every x ; in plain language, it keeps the learned residual update perpendicular to the fixed anchor. The router, residual subspaces, residual-coordinate maps, and adapter are then trained on a language-model or task loss. Held-out evaluation must compare against adapters with the same ranks and parameter count but no shared-anchor constraint.

VII. DISCUSSION

The chapter’s positive geometric thesis is the abundance–access distinction. At a fixed angular tolerance, high dimension permits an exponentially large finite dictionary, but a decoder still pays for the features that activate together. The corresponding semantic hypothesis is concrete: training should allocate lower task-weighted cross-talk to important, frequently coactive features than appears in isotropic and activity-matched controls.

The shared-anchor residual-subspace proposal adds a separate inductive bias for lexical ambiguity. Reusing one lexical ray across context classes should reduce sample complexity when the common component is real and per-use data are scarce. It does not add expressive power, and it should lose when forced sharing is wrong. Learning curves against equally expressive unstructured adapters or mixture models therefore test the proposed mechanism directly.

A. Relation to superposition and sparse autoencoders

The superposition hypothesis holds that neural networks may represent more features than they have neurons by using nonorthogonal directions with sparse activity [15, 16]. The null model separates dictionary coexistence from simultaneous linear readout. In the isotropic, weak-dependence model, many directions can coexist while a fixed-target correlation decoder accumulates interference of scale $\sqrt{k/d}$. Sparse activity controls that interference for this decoder; it is not asserted to be universally necessary for structured dictionaries or non-linear recovery.

Sparse autoencoders (SAEs) give a concrete empirical interface. They approximate layer activations as sparse expansions over learned decoder directions, making the w_i of Eq. (18) measurable objects rather than hypothetical features. Recent work has scaled SAEs, studied SAE feature geometry, and automated interpretation of large numbers of latents [17, 28–30]. In the present framework, SAE decoder directions make the Lévy/JL calibration and coactivation-weighted diagnostic measurable, while sense-conditioned SAE activation clusters can test the stronger shared-anchor residual-subspace hypothesis.

B. Relation to quantum cognition and compositional semantics

The phrase “quantum-inspired” has an established cognitive and linguistic lineage. Quantum-cognition models use Hilbert-space states, projectors, interference, and order effects as mathematical models of human judgment and concept combination, without requiring quantum processes in the brain [31, 32]. Work on the human mental lexicon has likewise used quantum structure to model word-association and memory experiments [33]. Those studies concern behavioral probabilities and conceptual judgments; the present chapter concerns hidden representations and learned readout in an LLM.

A different but neighboring tradition is compositional distributional semantics. Coecke, Sadrzadeh, and Clark combine vector-space word meanings with grammatical composition through tensor and categorical maps [34]. That construction shows how tensor products can organize composition, but it does not establish that a Transformer residual stream has a fully accessible semantic tensor factorization $d = q^n$.

The present model is a classical tied-subspace adapter: a router mixes context-conditioned residual subspaces that share one lexical ray. Its quantum-logic inspiration is exhausted by that incidence fact. It does not invoke noncommuting observables, order effects, complex amplitudes, unitary dynamics, or Born probabilities. Calling it quantum-inspired identifies the source of the structural constraint, not the physical nature of the computation.

C. Limitations

The limitations can be grouped into four issues.

Geometry. Lévy’s lemma controls a fixed Lipschitz probe; a finite dictionary additionally needs a union bound or code construction. JL gives an existence scale, the random-code formula gives an ensemble calibration, and the Welch bound gives a deterministic floor. None is a semantic upper bound.

Measurement protocol. The usable rank d and dictionary size M depend on layer choice, whitening, feature extraction, normalization, and duplicate removal. Contextual states can be anisotropic [35–38], although anisotropy is not universal [39]. Whitening changes the inner product and therefore changes angles. Finite-dimensional norm equivalence preserves convergence and continuity, but it does not preserve numerical angles or coherence. Every comparison must therefore use one declared metric and selection rule.

Readout. The law $\sigma_{\text{int}} \sim \sqrt{k/d}$ concerns one correlation decoder under random-like cancellation. Structured dictionaries, nonlinear decoders, inactive-feature comparisons, and correlated errors can change it. Likewise, Eq. (25) omits cross-moment terms unless the stated zero-cross-moment assumption holds.

Shared-anchor residual-subspace model. The model

may simply be wrong. Token identity can make its weak version pass while its nontrivial held-out comparison fails. Anchors, ranks, residual subspaces, and routers must be fixed on training data, and the true sense label is unavailable at test time. The proposed controls decide whether the tied model adds held-out value beyond ordinary token identity, clustering, and equally expressive untied adapters.

D. Proposed experiments and failure criteria

No experiment is reported in this chapter. The purpose of this section is to turn the analytical claims into a staged, falsifiable program. The first stage calibrates geometry; the later stages test increasingly restrictive semantic structure.

Geometric calibration (not a semantic claim). For a dictionary independently established to be overcomplete, compare its coherence, overlap distribution, and fixed-target readout error with isotropic and activity-matched controls. Agreement with Eq. (13) is a baseline result, not evidence that overcompleteness is useful; departure from it must be diagnosed before being called learned structure. JL itself is a theorem; the empirical question is whether task-relevant geometry and readout survive projection near its logarithmic dimension scale.

Claim A0: token-component sanity check. A token-associated component explains cross-context variation shared by occurrences of the same token and carries non-negligible held-out token information relative to matched random rays. This weak result is expected and is not evidence for a privileged semantic sector, but it prevents the strict model from passing with a zero or negligible anchor.

Claim A1: nontrivial shared-anchor claim. Relative to token-mean subtraction, top-principal-component removal, and same-rank random or supervised controls, a nondegenerate lexical ray $\mathcal{R}_t$ fixed from training data exposes held-out predictive information in $\mathcal{R}_t^\perp$ while its anchor coefficient remains sufficiently stable across contexts for a normalized tolerance fixed on training data. A0 is a prerequisite for A1.

Claim B: low-dimensional residual subspaces. Within $\mathcal{R}_t^\perp$, context-conditioned residuals admit a cross-validated low-rank description that predicts or compresses held-out states better than shuffled-label, ordinary clustering, and matched-subspace controls. Related polysemous senses need not have minimal cross-sense coherence.

Claim C: useful shared-anchor context classes. Without access to the true sense label at inference, a router and residual subspaces tied through the same lexical anchor improve held-out prediction, compression, interpretability, or sample efficiency relative to parameter-matched untied subspaces, unstructured low-rank adapters, and mixture-of-experts controls. The predicted advantage is largest for scarce or imbalanced

senses and may vanish when no stable shared component exists.

These claims can fail independently. The geometric calibration can hold while all structural claims fail. A0 may pass while A1 fails; that outcome would reject a privileged lexical anchor without denying ordinary token identity. Claim B can hold relative to a non-token-associated subspace, and A1 and B can hold even if the tying in Claim C adds no predictive value.

a. Minimum pilot. A small first study needs no retraining of the language model. Choose one public model, one layer, and a prespecified sample of ambiguous tokens from SemCor or the Word-in-Context (WiC) benchmark [40]. Estimate candidate token means and top principal components on the training split only. SemCor supports a linear sense classifier; for WiC use its intended pairwise same-sense versus different-sense task. Compare full states, residuals after each candidate removal, and same-rank random controls. Also report held-out anchor energy, token-identification information, and normalized coefficient variation. Together, held-out accuracy, calibration, nondegeneracy, and stability decide A1. Adding sparse-autoencoder features then tests the coactivation statistic. A failure at this stage should stop the tied-subspace study rather than be hidden by a more flexible router.

The following tests separate the possibilities.

1. *Lévy/random-code calibration.* Fix the carrier metric, spectral threshold or whitening rule, feature extractor, and duplicate criterion before evaluation. Report the selected rank d , dictionary size M , maximum coherence, the full overlap distribution, and active-set statistics. Compare with the Welch floor, isotropic dictionaries, and controls matched for marginal activity and coactivation. Treat Eq. (13) as a finite-sample null prediction, not a pass/fail capacity test. Also compare the signed pairwise distribution with the independently isotropic variance $1/d$; deviations may reflect dependence, anisotropy, extraction choices, or learned alignment.
2. *JL projection-stability test.* Fix a finite evaluation set of M normalized directions or contextual states, include the origin, and prespecify a relative squared-distance distortion δ_{JL} . Project to d_{proj} dimensions using frozen random maps scaled to preserve squared lengths on average, and measure pairwise distortion and held-out task performance as d_{proj} varies. JL predicts metric preservation when d_{proj} is proportional to $\delta_{\text{JL}}^{-2} \log(M+1)$ up to universal constants; preservation of semantic performance is the additional empirical question. Compare random Gaussian or orthogonal maps with learned low-rank maps at the same d_{proj} .
3. *Shared-anchor test.* For a lexically ambiguous token, choose candidate lexical rays generated by the

mean contextual embedding, a static input embedding explicitly mapped into the same layer carrier, or a learned token-specific direction. Fix the candidate on training data, then project held-out layer states with P_t and $I - P_t$ as in Eq. (33). Report the variation of the anchor coefficient $r_t^\top h_\ell(t | x)$ across contexts; exact constancy is the strict model, while any relaxed tolerance must be chosen before seeing test results. Sense labels should be better separated in $\mathcal{R}_t^\perp$ than in orthogonal complements of random, frequency-matched, or layer-matched control rays, and than in separately centered, token-mean-subtracted, or top-principal-component-removed baselines. Require the candidate anchor to carry prespecified nonnegligible token energy or token-identification information relative to matched random rays; report a normalized stability statistic such as $\text{Var}_x(r_t^\top h_\ell(t | x)) / \mathbb{E}_x \|h_\ell(t | x)\|^2$, where Var_x and $\mathbb{E}_x$ are the empirical variance and mean over held-out occurrences of the token. Compare held-out classification, compression, or a prespecified measure of added predictive information, with uncertainty estimated by label shuffling and resampling token instances.

4. *Operational residual-subspace test.* For each sense s , collect projected states $z_\ell(t, x) \in \mathcal{R}_t^\perp$ and estimate a low-rank subspace by principal component analysis (PCA) or probabilistic factor analysis. PCA is an application of the spectral theorem to the sample covariance matrix: its directions are eigenvectors with the largest eigenvalues. Select m_s without test-label leakage and use an orthonormal basis $B_{t,s}$. Evaluate held-out reconstruction and sense prediction against shuffled labels, ordinary supervised subspaces, and unsupervised clustering of equal rank. Report $\|B_{t,s}^\top B_{t,s'}\|_{\text{op}}$ descriptively, without assuming related senses must be nearly orthogonal.
5. *Anchor-versus-residual readout test.* For the relaxed diagnostic states, rather than the strict fitted model in Eq. (42), linear probes trained on $z_\ell(t, x)$ should perform comparably to probes trained on the full state $h_\ell(t | x)$, while probes restricted to the anchor component $P_t h_\ell(t | x)$ should provide less incremental sense information after frequency, syntax, and sense-prior controls. Near-chance performance is not required because the reference component may legitimately retain correlated information.
6. *Shared-anchor residual-subspace test.* Fit r_t , $B_{t,s}$, the residual-coordinate maps, and the router on training sequences, then evaluate Eqs. (40) and (41) without true sense labels on held-out instances. Compare with untied subspaces, unstructured low-rank adapters, ordinary contextual probes, parameter-matched mixture-of-

experts models, and shuffled subspace or sense labels. Repeat the comparison across training-set sizes and sense-imbalance levels. The tying mechanism predicts a larger advantage in the low-data regime and no guaranteed advantage when data are abundant or the shared component is weak.

7. *Synthetic concurrency degradation.* At a fixed layer, inject controlled superpositions

$$h_k = h_0 + \lambda \sum_{j=1}^k x_j w_j, \quad (43)$$

where h_0 is a held-out baseline activation, the w_j are independently chosen unit injection directions, the coefficients x_j are independent, centered, and unit-variance, and λ controls injection amplitude. The directions may come, for example, from SAE decoder directions or frozen linear probes. For random-like directions under the specified correlation decoder, the injected fixed-target interference has RMS scale $\lambda\sqrt{k/d}$, up to the distinction between k and $k-1$. How that interference changes downstream task performance is an empirical question. Include norm-preserving, random-direction, activation-frequency, and out-of-distribution controls; degradation by itself does not validate semantic superposition.

8. *Coactivation-dependent geometry.* At matched marginal activation frequency, feature similarity, SAE reconstruction error, amplitude, and task relevance, important frequently coactive features should contribute less to Eq. (25) than activity-matched controls. This tests learned redistribution of interference, independently of the shared-anchor hypothesis.

E. Scope of the quantum analogy

The complete implementation described in this chapter is classical: a router mixes real low-rank residual subspaces tied through one real lexical ray. Its quantum-logic inspiration consists of one structural fact. A projective measurement context can be represented by an orthonormal basis, or by the rank-one projectors associated with its basis vectors. Distinct contexts may contain the same projector. Figure 1 copies that shared-incidence pattern: several context hyperedges contain P_t .

The contrast is equally important. For a fixed quantum state ρ , the Born probability $\text{tr}(\rho P_t)$ of that shared projector does not depend on which compatible projectors complete the measurement context. In the proposed LLM model, the router probabilities, residual coordinates, and complete representation depend deliberately on the linguistic context x . The router is learned by ordinary optimization and obeys no Born or Gleason rule.

Noncommuting matrices, order effects, complex phases, unitary dynamics, and physical quantum states are not part of the proposal.

This limited probability consistency should not be paraphrased as saying that quantum mechanics is wholly noncontextual. Kochen–Specker results concern the impossibility of certain global noncontextual sharp-value assignments [41, 42]; no analogous theorem is claimed for semantic embeddings. The words “quantum-inspired” identify only the source of the shared-incidence constraint.

VIII. CONCLUSION

The central organizing distinction is between abundance and access. Lévy concentration, the Johnson–Lindenstrauss lemma, and random spherical coding are related expressions of the same high-dimensional concentration phenomenon: at fixed angular tolerance, a d -dimensional carrier admits exponentially many directions. The Welch bound gives a deterministic floor on their worst overlap. None of these counts makes the directions simultaneously readable. A simple readout from k random-like active features instead pays an RMS interference cost of order $\sqrt{k/d}$.

This distinction yields the chapter’s positive semantic prediction. A useful learned dictionary should not merely contain many directions; it should place lower task-weighted cross-talk between important features that often coactivate. Equation (25) turns that idea into a held-out comparison against isotropic and activity-matched controls.

For lexical ambiguity, the proposed tied adapter preserves a fixed lexical anchor while the surrounding text selects or mixes low-rank residual context subspaces. Sharing is predicted to improve sample efficiency when a genuine common component exists and per-sense data are scarce. The weak existence of a token component is trivial; the substantive test is incremental held-out value beyond token means, principal-component removal, and parameter-matched untied adapters.

The minimum pilot in Sec. VIID can reject the shared-anchor claim. The implementation is classical. Its only quantum-logic-inspired element is the incidence pattern of one shared projector in several orthonormal contexts; no Born-rule, noncommutativity, or semantic Kochen–Specker claim is made. The doubly exponential tensor-product count remains only the conditional remark of Sec. IIIC, not a conclusion about ordinary LLMs.

Appendix A: Explicit formalization of the three computational loops

The following subsections correspond one-to-one to Secs. IIA–IIC. The purpose is not to specify every engineering detail of a modern model. It is to identify, in

the order in which they occur, which objects are discrete symbols, which are vectors, which are temporary results of a calculation, and which are persistent model parameters. No argument about the semantic geometry in the main text depends on this appendix.

We first fix the objects used throughout. Let $\mathcal{S}$ be the set of finite character strings and let $\mathcal{V} = \{1, \dots, V\}$ be a finite vocabulary of V token identifiers. A token identifier is an integer label; it is not itself a vector. For a nonnegative integer T , the Cartesian power $\mathcal{V}^T$ is the set of token sequences of length T ; the union below allows every finite length. The tokenizer is a fixed, variable-length map

$$\text{Tok} : \mathcal{S} \longrightarrow \bigcup_{T \geq 0} \mathcal{V}^T. \quad (\text{A1})$$

Thus Tok turns one string into a finite sequence of identifiers. Its companion Detok assembles a token sequence into text. Apart from any declared normalization of spaces or characters, detokenization reverses the tokenization of an admissible string. Neither map learns a contextual meaning during the Transformer calculation.

The symbol d_{model} denotes the architectural width of the residual stream, so every hidden state below is an ordinary real column vector in $\mathbb{R}^{d_{\text{model}}}$. It must not be confused with the usable semantic carrier rank d in the main text: restriction, whitening, or removal of inactive directions may give $d \leq d_{\text{model}}$, with equality when the full residual stream is used. The symbol L denotes the number of Transformer layers. The symbol Θ denotes the complete collection of persistent numerical parameters: the token embedding table, the matrices in all attention and feed-forward blocks, normalization parameters, and the final vocabulary readout. If all their entries are written in one long list, then Θ may be regarded as a vector in $\mathbb{R}^{N_{\text{par}}}$, where N_{par} is the number of trainable scalar parameters. This flattening is only bookkeeping; an implementation stores the parameters as many arrays of different shapes. Thus d_{model} , V , and N_{par} count three different things: architectural hidden coordinates, token types, and scalar model parameters.

Persistent parameters survive from one prompt to the next. Hidden states, query, key, and value vectors, attention coefficients, and next-token probabilities are transient quantities recomputed for a particular input. The phrase “attention weight” is therefore potentially confusing: an attention coefficient is a temporary number, whereas a model weight is an entry of a persistent matrix in Θ . A superscript in parentheses denotes a stage or index rather than exponentiation: $h_i^{(\ell)}$ will index a layer, while $\Theta^{(r)}$ with $r = 0, 1, 2, \dots$ will index a successive parameter update.

1. Creating the weights

This subsection follows Sec. II A: text is tokenized, tokens are represented by vectors, self-attention makes

those vectors context-dependent, a next-token error is calculated, and an optimizer changes the persistent parameters.

a. Step 1: turn text into discrete identifiers. Consider one training string $\sigma \in \mathcal{S}$ whose selected training segment contains $T \geq 2$ tokens, so that it supplies at least one next-token prediction. Systems often use declared beginning- or end-of-sequence tokens at document boundaries. For $i \in \{1, \dots, T\}$, let $x_i \in \mathcal{V}$ denote the identifier at position i . Tokenization gives

$$(x_1, \dots, x_T) = \text{Tok}(\sigma) \in \mathcal{V}^T. \quad (\text{A2})$$

The same identifier may occur several times. At this stage all of its occurrences carry the same discrete label, irrespective of context.

b. Step 2: look up an initial vector and mark its position. Let $E_{\text{tok}} \in \mathbb{R}^{V \times d_{\text{model}}}$ be the token embedding table. It is one of the trainable arrays contained in Θ . Its row $E_{\text{tok}}[a] \in \mathbb{R}^{d_{\text{model}}}$ is the lookup vector for token type $a \in \mathcal{V}$; square brackets mean table lookup. In the simplest additive description, let $\rho_i^{\text{pos}} \in \mathbb{R}^{d_{\text{model}}}$ be a position vector. The initial hidden vector $h_i^{(0)} \in \mathbb{R}^{d_{\text{model}}}$ is

$$h_i^{(0)} = E_{\text{tok}}[x_i] + \rho_i^{\text{pos}}, \quad i = 1, \dots, T. \quad (\text{A3})$$

The superscript (0) means “before the first Transformer layer.” The position vector may be fixed or learned; when it is learned, its entries also belong to Θ . Systems using rotary or other positional schemes do not literally add ρ_i^{pos} , but they serve the same present purpose: the calculation must know both which token occurred and where it occurred. Two occurrences of “bank” begin with the same row of E_{tok} , but their later hidden states can differ because their surrounding tokens and positions differ.

c. Step 3: make the vectors context dependent by self-attention. We spell out one attention head in one layer. Let $h_i^{(\ell-1)} \in \mathbb{R}^{d_{\text{model}}}$ be the vector at position i entering layer $\ell \in \{1, \dots, L\}$. Let d_{head} be the width of one head, and let $W_Q^{(\ell)}, W_K^{(\ell)}, W_V^{(\ell)} \in \mathbb{R}^{d_{\text{head}} \times d_{\text{model}}}$ be three learned matrices contained in Θ . For target position i and source position j , write q_i, k_j, v_j for the three outputs

$$\begin{aligned} q_i &= W_Q^{(\ell)} h_i^{(\ell-1)}, \\ k_j &= W_K^{(\ell)} h_j^{(\ell-1)}, \\ v_j &= W_V^{(\ell)} h_j^{(\ell-1)}. \end{aligned} \quad (\text{A4})$$

The vectors $q_i, k_j, v_j \in \mathbb{R}^{d_{\text{head}}}$ are called the query, key, and value. A query describes what information position i is seeking, a key describes how position j may be matched, and a value contains the information to be combined if that position receives attention. These phrases interpret three ordinary matrix–vector products; they do not introduce new kinds of mathematical objects.

For a decoder-only model, position i may attend only to positions $j \leq i$. Let s_{ij} denote the real comparison

score, let α_{ij} denote the normalized attention coefficient, and let o_i^{att} denote the output of this head. The letter t below is only a dummy source-position index. These quantities are

$$\begin{aligned} s_{ij} &= \frac{q_i^\top k_j}{\sqrt{d_{\text{head}}}}, \\ \alpha_{ij} &= \frac{\exp(s_{ij})}{\sum_{t=1}^i \exp(s_{it})}, \\ o_i^{\text{att}} &= \sum_{j=1}^i \alpha_{ij} v_j. \end{aligned} \quad (\text{A5})$$

The first line uses the Euclidean inner product. By the Cauchy–Schwarz inequality, $|q_i^\top k_j| \leq \|q_i\| \|k_j\|$, so a large comparison requires suitably aligned query and key vectors as well as suitable lengths. The factor $1/\sqrt{d_{\text{head}}}$ controls the numerical scale of the scores. The ordinary exponential function is positive, hence every $\alpha_{ij} \geq 0$, and direct summation gives $\sum_{j=1}^i \alpha_{ij} = 1$. Consequently o_i^{att} , the output of this attention head, is a weighted average, or convex combination, of the permitted value vectors. The restriction $j \leq i$ is the causal mask: it prevents the calculation at position i from using the future target x_{i+1} .

A practical layer has several heads. Their outputs are concatenated, mapped back to $\mathbb{R}^{d_{\text{model}}}$, combined with a residual connection, normalized, and passed through a feed-forward block. Denote the resulting vector by $h_i^{(\ell)} \in \mathbb{R}^{d_{\text{model}}}$. Repeating the procedure for $\ell = 1, \dots, L$ yields final contextual vectors $h_i^{(L)}$. Let F_Θ denote the complete layered Transformer map. Then

$$(h_1^{(L)}, \dots, h_T^{(L)}) = F_\Theta(h_1^{(0)}, \dots, h_T^{(0)}) \quad (\text{A6})$$

Equation (A6) collects all these operations into one function. During this forward pass Θ is fixed. In contrast, $q_i, k_j, v_j, \alpha_{ij}$, and $h_i^{(\ell)}$ depend on the current string and are transient.

d. Step 4: turn the final vector into a next-token distribution. Let $W_{\text{out}} \in \mathbb{R}^{V \times d_{\text{model}}}$ and $b \in \mathbb{R}^V$ be the vocabulary readout matrix and bias; both belong to Θ . At training position i , let $g_i \in \mathbb{R}^V$ denote the score vector. For a candidate token $a \in \mathcal{V}$, let $p_\Theta(a \mid x_1, \dots, x_i)$ denote its predicted probability given the prefix. The letter c below is only a dummy token index in the denominator. Define

$$\begin{aligned} g_i &= W_{\text{out}} h_i^{(L)} + b, \\ p_\Theta(a \mid x_1, \dots, x_i) &= \frac{\exp(g_{i,a})}{\sum_{c \in \mathcal{V}} \exp(g_{i,c})}. \end{aligned} \quad (\text{A7})$$

The entry $g_{i,a}$ is a real-valued score, often called a logit, not a probability. The denominator is a finite sum of positive numbers, so the softmax expression is nonnegative and sums to one over $a \in \mathcal{V}$. The vertical bar means “given the prefix $(x_1, \dots, x_i)$.” Because of the causal mask, this distribution depends on that prefix and not on the unseen suffix $(x_{i+1}, \dots, x_T)$.

e. Step 5: compare the prediction with the observed continuation. Let $P_\Theta(x_2, \dots, x_T \mid x_1)$ denote the sequence-level conditional probability assigned to the observed continuation. The multiplication rule for conditional probabilities gives

$$P_\Theta(x_2, \dots, x_T \mid x_1) = \prod_{i=1}^{T-1} p_\Theta(x_{i+1} \mid x_1, \dots, x_i). \quad (\text{A8})$$

This product does not assume that tokens are independent: every factor is conditioned on the entire available prefix. Taking the negative natural logarithm turns the product into a sum. The loss for this string is

$$\mathcal{L}_\sigma(\Theta) = - \sum_{i=1}^{T-1} \log p_\Theta(x_{i+1} \mid x_1, \dots, x_i). \quad (\text{A9})$$

Each summand is small when the observed next token was assigned a probability near one and large when it was assigned a probability near zero.

f. Optional calculus detail: the initial error signal. The following derivative may be skipped without losing the computational sequence. Let ℓ_i be the i th summand in Eq. (A9), and let the indicator $\mathbf{1}_{\{a=x_{i+1}\}}$ equal one when a is the observed next token and zero otherwise. Differentiating the logarithm and softmax gives

$$\frac{\partial \ell_i}{\partial g_{i,a}} = p_\Theta(a \mid x_1, \dots, x_i) - \mathbf{1}_{\{a=x_{i+1}\}}. \quad (\text{A10})$$

Thus the derivative compares the predicted probability with a target that places all its mass on the observed token. This is the initial error signal propagated backward through the model.

g. The parameter update. Let $\mathcal{B}$ be a finite mini-batch of training strings and let $\hat{\mathcal{L}}_\mathcal{B}$ be the average of their string losses, usually normalized by the number of predicted tokens. The hat emphasizes that this is a finite-sample average. Let G denote its gradient with respect to all parameters, let $\eta > 0$ be a learning rate, and let Θ_{new} denote the parameters after one update. Then

$$\begin{aligned} G &= \nabla_\Theta \hat{\mathcal{L}}_\mathcal{B}(\Theta), \\ \Theta_{\text{new}} &= \Theta - \eta G, \quad \eta > 0. \end{aligned} \quad (\text{A11})$$

After flattening Θ into $\mathbb{R}^{N_{\text{par}}}$, the gradient ∇_Θ is the vector of partial derivatives with respect to every parameter, so G has the same number of scalar entries as Θ . Backpropagation computes it by repeatedly applying the multivariable chain rule through the readout, feed-forward blocks, and attention operations.

h. Optional calculus detail: why subtract the gradient? The symbol $\|G\|$ denotes the Euclidean length of G , and $\approx$ below means equality to first order in the small step size. The elementary first-order Taylor formula explains the minus sign: for a sufficiently small step and a differentiable loss,

$$\hat{\mathcal{L}}_\mathcal{B}(\Theta - \eta G) \approx \hat{\mathcal{L}}_\mathcal{B}(\Theta) - \eta \|G\|^2. \quad (\text{A12})$$

The omitted remainder becomes negligible compared with the linear term as $\eta \rightarrow 0$. Therefore a nonzero negative-gradient step locally lowers the same batch loss when the step is small enough; it does not promise a global minimum. Practical optimizers use running averages, adaptive scaling, regularization, and many batches, so they need not equal this simple update. The logical division remains: self-attention changes temporary activations, whereas the optimizer changes persistent parameters. Starting from initialized values and repeating the forward–loss–backward–update cycle produces the trained weights.

All spaces in this derivation are finite-dimensional Euclidean spaces. Calling $\mathbb{R}^{d_{\text{model}}}$ a real Hilbert space is formally correct because it is complete, but no completeness argument and no quantum postulate is needed. The linear algebra consists of vectors, matrices, inner products, and finite sums; the calculus consists mainly of differentiation and the chain rule.

2. Generating the output

This subsection follows Sec. IIB. The tokenizer and forward calculation are the same as during training, but the observed next token is now unavailable. The model must choose one token, append it, and use the enlarged prefix for the next choice.

a. Initial state of the loop. Let $\sigma_{\text{prompt}} \in \mathcal{S}$ be the complete prompt, including any visible role or control tokens. Suppose tokenization returns m identifiers, where m is the prompt length, and let $u_i \in \mathcal{V}$ be the identifier at prompt position i . Then

$$(u_1, \dots, u_m) = \text{Tok}(\sigma_{\text{prompt}}) \in \mathcal{V}^m. \quad (\text{A13})$$

Each u_i is an integer token identifier, not a vector. We use u rather than the training symbol x only to keep the two discussions visually separate. Let Θ^* denote the deployed parameter values obtained after training and any declared post-training. The star labels the deployed checkpoint; it does not denote complex conjugation. Ordinary generation holds Θ^* fixed.

Apply the lookup and Transformer operations in Eqs. (A3)–(A6) to the prompt. After the prefix has length $j \geq m$, let $h_j^{(L)}$ be its last contextual vector and let $g_j = W_{\text{out}} h_j^{(L)} + b \in \mathbb{R}^V$ be the score vector. Its entry $g_{j,a}$ is the score for candidate token $a \in \mathcal{V}$.

b. Convert the scores into a decoding rule. Choose a positive decoding temperature τ_{dec} . Let $\pi_j(a)$ denote the normalized probability of candidate token a , and let $u_{1:j}$ abbreviate the prefix $(u_1, \dots, u_j)$. The index c below is a dummy token index. The notation $u_{j+1} \sim \pi_j$ will mean that the next token identifier is sampled from π_j ; here ~

means sampling, not approximate equality. Define

$$\begin{aligned} \pi_j(a) &= \frac{\exp(g_{j,a}/\tau_{\text{dec}})}{\sum_{c \in \mathcal{V}} \exp(g_{j,c}/\tau_{\text{dec}})}, \quad a \in \mathcal{V}, \\ u_{j+1} &\sim \pi_j, \\ u_{1:j+1} &= (u_{1:j}, u_{j+1}). \end{aligned} \quad (\text{A14})$$

The parentheses in the last line mean sequence concatenation. The function $\pi_j : \mathcal{V} \rightarrow [0, 1]$ is a probability distribution because its entries are nonnegative and its denominator makes $\sum_{a \in \mathcal{V}} \pi_j(a) = 1$.

Thus a token identifier is not being equated with a probability. Score differences determine probability ratios:

$$\frac{\pi_j(a)}{\pi_j(c)} = \exp\left(\frac{g_{j,a} - g_{j,c}}{\tau_{\text{dec}}}\right). \quad (\text{A15})$$

A smaller positive temperature magnifies score differences and makes the distribution more concentrated; a larger temperature flattens it. In greedy decoding, sampling is replaced by $u_{j+1} = \arg \max_{a \in \mathcal{V}} g_{j,a}$, meaning that the candidate token with the largest score is selected. Other rules may retain only a declared subset of candidates and then renormalize. These are decoding choices, not additional training steps.

c. Feed the chosen token back as part of the next input. The last line of Eq. (A14) appends u_{j+1} to the prefix. The enlarged sequence is then passed through the same fixed-parameter calculation to obtain g_{j+1} and the following distribution. This is a recurrence: the state used at one step contains the output of the preceding step. It is not gradient descent because no derivative is taken and Θ^* does not change. A token selected early in the response can alter all later query–key comparisons and therefore all later distributions.

Let J be the final sequence position and let $u_{m+1:J}$ abbreviate the complete generated continuation $(u_{m+1}, \dots, u_J)$. Write $\mathbb{P}_{\Theta^*, \tau_{\text{dec}}}$ for the probability assigned by the fixed deployed model and the displayed temperature-sampling rule. The multiplication rule gives

$$\mathbb{P}_{\Theta^*, \tau_{\text{dec}}}(u_{m+1:J} \mid u_{1:m}) = \prod_{j=m}^{J-1} \pi_j(u_{j+1}). \quad (\text{A16})$$

This product does not assert independence. Each factor π_j was calculated from the complete preceding prefix $u_{1:j}$. Generation may stop when a declared end token is chosen, a length limit is reached, or another stopping rule fires. The newly generated answer is

$$\sigma_{\text{answer}} = \text{Detok}(u_{m+1}, \dots, u_J). \quad (\text{A17})$$

The prompt tokens are omitted because this expression denotes only the new answer text.

An implementation usually stores previously computed key and value vectors in a key–value cache. The cache avoids recomputing the same transient quantities for the

unchanged prefix. It changes time and memory use, but it is neither an external factual memory nor a change to the parameters Θ^* . In summary, the token sequence, attention coefficients, and hidden vectors change at every generation step; the deployed model weights do not.

3. Recursive adaptation, alignment, and hallucination

This subsection expands Sec. II C. The word “recursive” can refer to three different loops, and only the third changes model weights.

a. Loop 1: autoregressive state recursion. Let $c_j = u_{1:j}$ denote the token prefix at generation step j . The update $c_{j+1} = (c_j, u_{j+1})$ changes the input state, while the deployed parameter vector remains Θ^* at every token step. This is precisely the generation loop in Eq. (A14). It supplies new context without learning new persistent parameters.

b. Loop 2: external memory or retrieval. Let $r = 0, 1, 2, \dots$ count complete interaction rounds, let I_r and O_r be the user’s input and the model’s output in round r , and let $\mathcal{M}$ be the set of permitted external-memory records. The symbol $m_r \in \mathcal{M}$ denotes the record after round r and is unrelated to the prompt-length integer m in the preceding subsection. Let U_{mem} be a rule that revises the record, let A_{mem} be a rule that assembles retrieved material into a prompt, and let $\sigma_{\text{prompt}}^{(r+1)}$ denote the resulting prompt for the next round. A memory system may use maps of the schematic form

$$\begin{aligned} m_{r+1} &= U_{\text{mem}}(m_r, I_r, O_r), \\ \sigma_{\text{prompt}}^{(r+1)} &= A_{\text{mem}}(I_{r+1}, m_{r+1}). \end{aligned} \quad (\text{A18})$$

The first line updates the external record; the second uses that record when constructing the next input string. These maps may be implemented by a database, a retrieval system, or a longer conversation history; they need not be linear. They alter what the model reads but still leave Θ^* fixed. This is why a key–value cache, a retrieved document, and a trained parameter are three different kinds of storage.

c. Loop 3: genuine online parameter adaptation. Let $\Theta^{(r)} \in \mathbb{R}^{N_{\text{par}}}$ denote the parameter vector available at the beginning of interaction round r . The starting value $\Theta^{(0)}$ may already include instruction or preference-based post-training. Let $\mathcal{B}_r$ be a finite collection of externally verified input–target examples approved for adaptation after round r , and let $\mathcal{J}_r(\Theta; \mathcal{B}_r)$ be the scalar loss chosen for those examples. Let $g_r \in \mathbb{R}^{N_{\text{par}}}$ denote the gradient of that loss at the current parameters. Thus

$$g_r = \nabla_{\Theta} \mathcal{J}_r(\Theta^{(r)}; \mathcal{B}_r) \in \mathbb{R}^{N_{\text{par}}}. \quad (\text{A19})$$

Thus g_r lists the derivative of the adaptation loss with respect to every scalar model parameter.

To state which parameters may change, let $\mathcal{P}_r^{(\Theta)} : \mathbb{R}^{N_{\text{par}}} \rightarrow \mathbb{R}^{N_{\text{par}}}$ be a declared linear mask or projection.

Choose a learning rate $\eta_r > 0$, and let $\Theta^{(r+1)}$ denote the parameter vector after the update. A restricted gradient step is

$$\Theta^{(r+1)} = \Theta^{(r)} - \eta_r \mathcal{P}_r^{(\Theta)} g_r, \quad \eta_r > 0. \quad (\text{A20})$$

In words: the new parameters equal the old parameters minus a chosen step size times the permitted part of the adaptation gradient.

If $\mathcal{P}_r^{(\Theta)} = 0$, no parameter changes. If it keeps only adapter coordinates, only that adapter changes. If it is the identity map, every parameter direction is permitted.

d. Optional linear-algebra detail: restricted descent. When $\mathcal{P}_r^{(\Theta)}$ is an orthogonal projection, it satisfies $(\mathcal{P}_r^{(\Theta)})^\top = \mathcal{P}_r^{(\Theta)}$ and $(\mathcal{P}_r^{(\Theta)})^2 = \mathcal{P}_r^{(\Theta)}$. Elementary linear algebra then splits g_r into an allowed part $\mathcal{P}_r^{(\Theta)} g_r$ and a perpendicular forbidden part. For a sufficiently small step, the same first-order argument as in Eq. (A12) gives

$$\mathcal{J}_r(\Theta^{(r+1)}; \mathcal{B}_r) \approx \mathcal{J}_r(\Theta^{(r)}; \mathcal{B}_r) - \eta_r \|\mathcal{P}_r^{(\Theta)} g_r\|^2. \quad (\text{A21})$$

The equality $g_r^\top \mathcal{P}_r^{(\Theta)} g_r = \|\mathcal{P}_r^{(\Theta)} g_r\|^2$ follows from symmetry and idempotence. If $\mathcal{P}_r^{(\Theta)} g_r \neq 0$, this calculation explains strict local descent on the declared adaptation loss for a sufficiently small step. It does not guarantee preservation of earlier abilities.

The superscript (Θ) reminds us that this operation takes place in the large parameter space. It is unrelated to the token-associated representation-space projector P_t in Sec. V. Equation (A20) restricts an update direction but does not by itself bound the update size. Clipping, a trust region, or an explicit norm bound can limit one step. A general projected-gradient algorithm may additionally project the updated point back onto a feasible set; that is a further operation not displayed here.

A user interaction normally supplies no verified target. Unrestricted updates can cause forgetting, distribution drift, privacy leakage, or data poisoning, and repeated training on unverified model output can reinforce errors [43]. A defensible online learner therefore needs provenance, external verification, bounded or adapter-local updates, held-out tests, and a recoverable reference model.

e. Alignment is a separate parameter intervention. Let Θ_{base} denote a fixed base model before a declared alignment procedure, and let $\mathcal{D}_{\text{align}}$ be the data used by that procedure. Let Align denote the complete post-training procedure and let Θ_{align} denote its resulting parameters. Schematically,

$$\Theta_{\text{align}} = \text{Align}(\Theta_{\text{base}}; \mathcal{D}_{\text{align}}). \quad (\text{A22})$$

The procedure need not be a linear map. Alignment therefore changes persistent weights, but it usually occurs before deployment rather than after every user interaction. The difference $\Theta_{\text{align}} - \Theta_{\text{base}}$ records a parameter change; its existence or length alone does not determine whether factual accuracy improves or worsens. That direction of effect must be measured against a matched control.

f. Define the measurement before comparing models.

Let q denote one evaluation case, including its prompt and any grounding material supplied to the model, and let $\mathcal{D}_{\text{eval}}$ be the fixed distribution of such cases. Let Y denote the generated response under one fixed decoding rule. Finally, let $Z(q, Y) \in \{0, 1\}$ be a prespecified response-level indicator: it is one if the response contains an unsupported claim under the declared judging rule and zero otherwise. In the next display, $\mathbb{E}$ means an average, each $\sim$ means “sampled from,” and $P_{\Theta}(\cdot | q)$ is the response distribution of model Θ under the fixed decoder; the dot is a placeholder for all possible responses. The population unsupported-response rate is

$$H(\Theta) = \mathbb{E}_{\substack{q \sim \mathcal{D}_{\text{eval}} \\ Y \sim P_{\Theta}(\cdot | q)}} [Z(q, Y)]. \quad (\text{A23})$$

In words: draw an evaluation case, generate a response, record either zero or one according to the judging rule, and average over repetitions. The probability law $P_{\Theta}(\cdot | q)$ includes the fixed decoder and its randomness. For deterministic greedy decoding it is concentrated on one response. A study that instead counts individual claims must redefine Z and must not treat several claims from the same response as independent cases.

For N held-out evaluation cases $q_1, \dots, q_N$, let n index a case and let $Y_n^{(\Theta)}$ be the response generated by model Θ on that case. Define the measured rate $H_N(\Theta)$ to be the ordinary sample mean

$$H_N(\Theta) = \frac{1}{N} \sum_{n=1}^N Z(q_n, Y_n^{(\Theta)}). \quad (\text{A24})$$

Every symbol in this quantity now has a defined empirical role: N counts evaluation cases, each Z is zero or one, and $H_N \in [0, 1]$ is their measured fraction.

For aligned parameters Θ_{align} and matched control parameters Θ_{ctrl} , let $\Delta_{\text{hall}}^{\text{align}}$ denote the population difference and let its hatted version denote the finite-sample estimate. The superscript “align” is a descriptive label, not a power. Define

$$\begin{aligned} \Delta_{\text{hall}}^{\text{align}} &= H(\Theta_{\text{align}}) - H(\Theta_{\text{ctrl}}), \\ \hat{\Delta}_{\text{hall}}^{\text{align}} &= H_N(\Theta_{\text{align}}) - H_N(\Theta_{\text{ctrl}}). \end{aligned} \quad (\text{A25})$$

The control Θ_{ctrl} must start from the same Θ_{base} and be matched as closely as possible in data exposure, optimization and compute budget, evaluation prompts, grounding, decoding, and random seeds, except for the declared alignment intervention. Only when that intervention is the sole systematic difference can the contrast be attributed causally to it. One may set $\Theta_{\text{ctrl}} = \Theta_{\text{base}}$ only when “no post-training” is the explicit control condition. If the control receives a different form of post-training, that procedure and its data must be stated.

The hypothesis that alignment increases hallucination in the declared regime is $\Delta_{\text{hall}}^{\text{align}} > 0$; reduction corresponds to a negative value, and zero means no population difference on this measure. Nothing in the formalism fixes the sign. Coverage and abstention must be reported beside H_N , since universal refusal could lower the unsupported-response rate without adding knowledge. If the joint pairs of evaluation cases and generated responses, including decoder randomness, are independent draws from the fixed protocol, the law of large numbers explains why H_N approaches H as N grows. A finite study still requires uncertainty intervals and variation across training seeds. The equations therefore turn the proposed concern about alignment into a testable comparison; they do not assume its conclusion.

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